

Natural Language Processing - Endsem

Full Marks: 95, Time : 3 hours, Date: 27.11.2025

Instructions: Don't mark any options in section 1, and don't fill in the blanks in section 2 of the question paper. All the questions from a section should be answered together. No doubts will be entertained.

1 MCQ [Full Marks: 20]

- (i) What is the main innovation of the transformer architecture compared to previous neural network models?
 - (a) Recurrent connections
 - (b) Convolutional layers
 - (c) Attention mechanism **Answer**
 - (d) Pooling layers
- (ii) What is the role of the residual stream in a transformer block?
 - (a) It stores the output tokens
 - (b) It passes and modifies token representations between layers **Answer**
 - (c) It computes the attention weights
 - (d) It generates the vocabulary
- (iii) Why is masking used in the self-attention mechanism for language modeling?
 - (a) To prevent overfitting
 - (b) To ignore padding tokens
 - (c) To prevent attending to future tokens **Answer**
 - (d) To speed up computation
- (iv) A certain news website allows readers to post anonymous comments on any article. The publisher hires you to write some software that will identify "good comments." The publisher plans to delete the bad comments automatically. The publisher insists that you must achieve 90% precision on the task of identifying good comments. What does that mean?
 - (a) Be nice to writers. It is okay if we accidentally delete a few of the good comments, but no more than 10% of them.
 - (b) Be nice to readers. It is okay if a few of the comments that appear on our website are bad, but no more than 10% of them. **Answer**
 - (c) Be accurate. It is okay to make some mistakes and misclassify a few comments, but no more than 10% of them.
 - (d) None of them.
- (v) Which technique is used to predict the probability of a sequence of words in a given context?
 - (a) Prompting
 - (b) Machine translation
 - (c) Language modeling **Answer**
 - (d) Named entity recognition
- (vi) Which metric is commonly used to evaluate machine translation systems?
 - (a) F1 score
 - (b) Precision
 - (c) Recall
 - (d) BLEU score **Answer**
- (vii) What is the purpose of the attention mechanism in neural networks?
 - (a) To improve computational efficiency
 - (b) To focus on relevant information **Answer**
 - (c) To reduce overfitting
 - (d) To calculate feature importance
- (viii) Which of the following is not a sequence labeling task?
 - (a) Named entity recognition
 - (b) Part-of-speech tagging
 - (c) Sentiment analysis **Answer**
 - (d) Chunking
- (ix) Which technique is used to improve the performance of machine translation models by leveraging monolingual data?
 - (a) Data augmentation
 - (b) Beam search
 - (c) Attention
 - (d) Backtranslation **Answer**
- (x) What is the primary objective of the alignment process in MT?
 - (a) Detect syntax errors
 - (b) Identify word or phrase correspondences between source and target texts **Answer**

- (c) Assign part-of-speech tags
(d) Encode grammar rules
- (xi) Which fine-tuning technique involves training only a small subset of additional parameters, like those in adapters or LoRA matrices, rather than updating the entire model?
(a) Standard fine-tuning
(b) Parameter-efficient fine-tuning (PEFT) **Answer**
(c) Task-specific fine-tuning
(d) Instruction set fine-tuning
- (xii) In supervised fine-tuning, what is typically required to optimize a model for task-specific performance?
(a) Large corpus of unlabeled data
(b) High-quality labeled datasets for each task **Answer**
(c) Only pre-trained language model
(d) Hyperparameter search
- (xiii) What distinguishes in-context learning from traditional supervised fine-tuning in NLP?
(a) It updates the model's all internal weights during inference
(b) It updates few internal weights during inference
(c) It uses task demonstrations in the prompt without altering model parameters **Answer**
(d) It requires large labeled datasets for each new task
- (xiv) What is the main function of the retriever component in a RAG system?
(a) Generating responses by synthesizing information
(b) Fine-tuning the language model
(c) Translating user queries into different languages
(d) Fetching relevant documents or data from knowledge sources **Answer**
- (xv) Why are RAG models preferred over standard language models for up-to-date factual question answering?
(a) They integrate real-time retrieval from external sources **Answer**
(b) They generate more creative stories
(c) They rely only on pre-trained knowledge
(d) They are less complex
- (xvi) Which metric is commonly used to evaluate the relevance and quality of retrieved documents in RAG systems?
(a) Mean average precision **Answer**
(b) Simple Precision-Recall
(c) F1-Score
(d) BLEU
- (xvii) What challenge does RAG address to reduce hallucinations in generated answers?
(a) Limiting answer length
(b) Only using training data
(c) By grounding outputs in retrieved real-world information **Answer**
(d) Increasing prompt diversity
- (xviii) Which of the following is a benefit of using chain-of-thought (CoT) prompting?
(a) It always produces shorter model outputs
(b) It enables the model to generate step-by-step reasoning for complex tasks **Answer**
(c) It improves image captioning specifically
(d) It restricts the model to only factual answers
- (xix) What does "zero-shot" prompting mean in the context of large language models?
(a) The model is trained on examples
(b) The task is performed with instructions only, no task-specific examples **Answer**
(c) The model generates random answers
(d) The entire dataset is fed to the model
- (xx) Which of the following statements best describes a masked language model (MLM)?
(a) It predicts the next word in a sequence based only on preceding words.
(b) It predicts masked or missing words in a sentence using both left and right context. **Answer**
(c) It generates text by predicting the next sentence.
(d) It predicts sentiment from a given text.

2 Fill in the Blanks [Full Marks=15X1=15]

- (i) The process of breaking running text into individual tokens, such as words or punctuation, is known as _____. **Answer: tokenization**

- (ii) Adding a penalty term to the loss/objective function to reduce overfitting is called _____. **Answer: regularization.**
- (iii) The proportion of correctly predicted positive items relative to all actual positives is called _____. **Answer: recall.**
- (iv) The BERT model uses _____ self-attention, allowing tokens to consider both preceding and following context. **Answer: bidirectional.**
- (v) The metric "Mean Reciprocal Rank" evaluates the ranking of the _____. **Answer: first.**
- (vi) The process of mapping multiple inflected forms to their dictionary form is called _____. **Answer: lemmatization/ stemming.**
- (vii) The component in QA that extracts the exact answer span from the retrieved text is the _____. **Answer: reader.**
- (viii) Subword tokenization helps mitigate the problem of _____. **Answer: Large vocabulary size.**
- (ix) In sequence labeling, the goal is to assign _____ to individual words or elements in a sequence. **Answer: Tags/labels.**
- (x) The probability of a tag following another tag in HMMs is the _____ probability. **Answer: Transition.**
- (xi) In transformer language modeling, the matrix that maps logits back to vocabulary probabilities is sometimes called the _____ layer. **Answer: unembedding.**
- (xii) "In-context learning" refers to a model learning a task from its _____ rather than from gradient updates. **Answer: prompts.**
- (xiii) Model alignment can be improved by fine-tuning LLMs on human-written instructions and using _____ learning from human feedback. **Answer: reinforcement.**
- (xiv) The set of sentences in two languages aligned line by line for training MT models is called a _____ corpus. **Answer: parallel/bitext**
- (xv) Languages in which pronouns can be omitted without loss of meaning, like Japanese, are called _____ languages. **Answer: pro-drop**

3 True-False with one line justification; no marks will be awarded if justification is not correct or long [Full Marks 2x10=20]

- (i) Positional encoding is unnecessary in transformer architectures because self-attention can capture sequence order directly.
Answer False, Justification: Transformer models require positional encoding to inject information about the order of tokens in a sequence, as self-attention alone cannot represent the sequence's order, unlike RNNs.
- (ii) The multi-head attention mechanism in transformers creates multiple sets of queries, keys, and values to attend to information from different representation subspaces.
Answer, True Justification: Multi-head attention enables the model to focus on different positions by having multiple sets of queries, keys, and values that learn various relationships within the input data.
- (iii) Retrieval-Augmented Generation (RAG) uses information retrieval techniques to find relevant documents before generating an answer.
Answer True RAG combines retrieval (finding texts likely to contain relevant facts) with a language model that generates answers based on those retrieved texts.
- (iv) TF-IDF and BM25 are term weighting schemes that treat all word occurrences equally in scoring.
Answer: False Both TF-IDF and BM25 consider term frequency and document frequency, and BM25 also normalizes for document length, so not all occurrences are weighted equally.
- (v) An inverted index allows fast retrieval of documents containing specific query terms and stores term frequencies.
Answer: True The inverted index efficiently maps terms to lists of documents that contain them, storing frequencies and other useful statistics.
- (vi) All natural languages have the same word order, which simplifies automatic translation.

tion.

Answer:False Languages differ widely in word order (e.g., SVO, SOV, VSO), and these differences are fundamental translation challenges.

- (vii) The chrF metric for evaluating MT is based on word n-gram overlap and averages both precision and recall.

Answer: False chrF assesses MT quality by measuring overlap in character-based n-grams, incorporating both precision and recall.

- (viii) BLEU, an MT evaluation metric, combines both precision and recall through the geometric mean.

Answer False BLEU is primarily a precision-based metric and does not combine recall with the geometric mean (recall is not included in BLEU).

- (ix) The raw dot product between two word vectors does not account for vector length and overly favors frequent words.

Answer True Without normalization, the dot product is higher for long vectors, which are typical of frequent words, making it a poor similarity metric by itself.

- (x) The Viterbi algorithm allows efficient inference by decomposing the scoring function into a sum of local parts.

Answer True The Viterbi algorithm takes advantage of this decomposition to perform efficient dynamic programming for sequence labeling.

4 Numericals[Full Marks:40]

- (i) Build the inverted index for the following corpus. It should include document frequency and term frequency. [4 marks].

D1	"AI is the future"
D2	"AI and machine learning"
D3	"machine learning is fun with learning"
D4	"deep learning shapes the future"

Answer

ai{2} → D1(1)– >D2(1)
 is{2} → D1(1)– >D3(1)
 the{2} → D1(1)– >D4(1)
 future{2} → D1(1)– >D4(1)
 and{1} → D2(1)

machine{2} → D2(1)– >D3(1)
 learning{3} → D2(1)– >D3(2)– >D4
 fun{1} → D3(1)
 with{1} → D3(1)
 deep{1} → D4(1)
 shapes{1} → D4(1)

- (ii) Given $Q=[1 \ 0 \ 1]$, $K=\begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$,
 $V=\begin{bmatrix} 1 & 0 \\ 0 & 2 \\ 1 & 1 \end{bmatrix}$ and $d_k=3$. Compute the attention output A. [3 Marks]

Answer

$$QK^T=[1 \ 0 \ 1]\begin{bmatrix} 1 & 0 & 1 \\ 2 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}=[2 \ 0 \ 2].$$

$$\frac{QK^T}{\sqrt{3}}=[1.155 \ 0 \ 1.155]$$

$$\text{softmax}([1.155 \ 0 \ 1.155])=\left[\frac{e^{1.155}}{e^{1.155}+e^0+e^{1.155}}, \frac{e^0}{e^{1.155}+e^0+e^{1.155}}, \frac{e^{1.155}}{e^{1.155}+e^0+e^{1.155}}\right]$$

$$=\left[\frac{3.174}{3.174+1+3.174}, \frac{1}{3.174+1+3.174}, \frac{3.174}{3.174+1+3.174}\right]$$

$$=[0.432, 0.136, 0.432]$$

$$[0.432, 0.136, 0.432]\begin{bmatrix} 1 & 0 \\ 0 & 2 \\ 1 & 1 \end{bmatrix}=[0.864 \ 0.704]$$

- (iii) A company wants to build a spam filter using the Naïve Bayes text classification model. The filter learns word probabilities from labeled emails (spam or not spam). When a new email arrives, the model predicts whether it's spam or ham (not spam). We have a simplified vocabulary containing only three words: ["offer", "buy", "meeting"]

From training data, we estimated the following probabilities:

Word	P(word spam)	P(word ham)
offer	0.5	0.1
buy	0.4	0.1
meeting	0.1	0.8

And class priors: P(spam)=0.6, P(ham)=0.4
 Now, a new email comes in with the words: "buy offer"

Determine whether it is spam or ham using the Naïve Bayes classifier [4 Marks].

Answer: We assume Naïve Bayes independence — that the probabilities of words

multiply independently.

For spam: $P(\text{buy,offer}|\text{spam})=P(\text{buy}|\text{spam}) \times P(\text{offer}|\text{spam}) = 0.4 \times 0.5 = 0.20$
 $P(\text{spam,buy,offer})=P(\text{spam}) \times P(\text{buy,offer}|\text{spam}) = 0.6 \times 0.20 = 0.12$
 For ham: $P(\text{buy,offer}|\text{ham}) = P(\text{buy}|\text{ham}) \times P(\text{offer}|\text{ham}) = 0.1 \times 0.1 = 0.01$
 $P(\text{ham,buy,offer}) = P(\text{ham}) \times P(\text{buy,offer}|\text{ham}) = 0.4 \times 0.01 = 0.004$
 So, it's a spam.

- (iv) A bigram language model is trained on a small text corpus of prior sentences. We have 4 training sentences (with start $< s >$ and end $< /s >$):

$< s > \text{ I love NLP } < /s >$

$< s > \text{ I love machine learning } < /s >$

$< s > \text{ I enjoy learning } < /s >$

$< s > \text{ I love coding } < /s >$

Compute relevant **smoothed** (Laplace smoothing) bigram probabilities and then find out the probability of sentence "*I love learning*" and also compare which is more probable sentence out of "*I love learning*" and "*I love NLP*"
[2+3+2=7 Marks]

Answer $P(\text{I} | < s >) = \frac{4+1}{4+9} = \frac{5}{13} = 0.3846$

$p(\text{love}|\text{I}) = \frac{3+1}{4+9} = \frac{4}{13} = 0.3077$

$p(\text{learning}|\text{love}) = \frac{0+1}{3+9} = \frac{1}{12} = 0.0833$

$p(< /s > | \text{learning}) = \frac{2+1}{2+9} = \frac{3}{11} = 0.2727$

$P(\text{NLP}|\text{love}) = \frac{1+1}{3+9} = \frac{2}{12} = 0.1667$

$P(\text{i/s}_i - \text{NLP}) = 1 + \frac{1}{1+9} = \frac{2}{10} = 0.2$

$p(< s > \text{ I love NLP } < /s >) = 0.3846 \times 0.3077 \times 0.0833 \times 0.2727 = 0.00269$

$P(< s > \text{ I love NLP } < /s >) = 0.3846 \times 0.3077 \times 0.1667 \times 0.2 = 0.00395$

I love NLP is more probable.

- (v) You are given a sentence and your goal is to assign part-of-speech tag corresponding to each words in sentence. Assume there are only two part-of-speech: Noun (N) and verb(V). You are also given transition score as well as emission score. Show the diagram of trellis. **[5+2=7 Marks]**

Transition Score:

From → To	N	V	End-Tag
Start-Tag	4	2	—
N	1	3	2
V	3	1	4

Emission Scores:

Word	N	V
Dogs	5	1
bark	1	4

Answer: Step 1: Initialization For first word
 Dogs:

$\text{Score(N)} = \text{Start} \rightarrow \text{N} + \text{Emission(Dogs, N)}$
 $= 4 + 5 = 9$

$\text{Score(V)} = \text{Start} \rightarrow \text{V} + \text{Emission(Dogs, V)}$
 $= 2 + 1 = 3$

Trellis at $t=1$: $N = 9, V = 3$

Step 2: Recursion (Word: "bark") We calculate scores for each current tag by combining previous path + transition score + emission score.

For N:

From N: $9 + 1 + 1 = 11$

From V: $3 + 3 + 1 = 7$ Max = 11 (path via N)

For V:

From N: $9 + 3 + 4 = 16$

From V: $3 + 1 + 4 = 8$ Max = 16 (path via N)

Trellis at $t=2$: $N = 11, V = 16$

Step 3: Termination Add transition to End tag:

From N: $11 + 2 = 13$

From V: $16 + 4 = 20$

Highest = 20 (from V), so best ending tag = V.

Step 4: Backtrace Since V (from "bark") came via N (from "Dogs"), the optimal path is:

Start $\rightarrow$ N $\rightarrow$ V $\rightarrow$ End

- (vi) Suppose an English-to-French neural translation model generates probabilities for three candidate French words at each decoding step. The goal is to translate the English word:
 "Go"

The target (French) vocabulary is limited to:

$\{ \text{Va, Vas, Allez, } < eos > \}$ Here $< eos >$ denotes the end-of-sequence token. The model will decode up to three steps, and we will use a beam width (k) = 2. Token probabilities at timestep $t=1$ and $t=2$ is given.

Step 1: Initial Token Probabilities (at $t = 1$)

Token	$P(\text{token} \text{context})$
Va	0.5
Vas	0.3
Allez	0.2

Step 2: Expansion ($t = 2$)

For each surviving sequence, the model predicts the next word probabilities.

Previous token	Next token	P(next previous)
Va	< eos >	0.6
Va	Vas	0.2
Va	Allez	0.2
Vas	< eos >	0.5
Vas	Va	0.3
Vas	Allez	0.2
Allez	Va	0.2
Allez	Vas	0.3
Allez	< eos >	0.4

“coffee” should be placed properly to show alignment. [8+4=12 Marks]

e							
f							
f							
a							
r							
i							
g							
		c	o	f	f	e	e

What will be the final translation? explain with beam search picture. [3 Marks]

- (vii) Show the edit distance between “giraffe” and “coffee” assuming insertion and deletion cost is 1 while substitution cost is 2. Fill up the table (format of the table given below). Suggest one possible alignment along with the operation applied at each position (i for insertion/ d for deletion/ s for substitution) for achieving optimized edit distance. For showing alignment, write “giraffe” on top and place “coffee” at the bottom, and the starting letter of

Answer

e	7	8	9	8	7	6	7
f	6	7	8	7	6	7	8
f	5	6	7	6	7	8	9
a	4	5	6	7	8	9	10
r	3	4	5	6	7	8	9
i	2	3	4	5	6	7	8
g	1	2	3	4	5	6	7
0	0	1	2	3	4	5	6
		c	o	f	f	e	e

One possible alignment

g i r a f f e
c o f f e e
d d s s i